Generalized Coordinates in RxInfer.jl

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All right, this is a short research update August 8th, 2025 about using generalized coordinates of motion models of various orders in RX infer to take noisy samples and do path inference on the underlying smooth trajectory and then compare how the different orders, so different moments of the distribution, how well they do in terms of confidence interval coverage. free energy minimization etc across different scenarios. So this is all in the rx infer examples.jl repo on my personal doxology fork. If you want to learn more about generalized coordinates start with the Wikipedia or the paper basian mechanics for stationary processes which is live stream 26. So here we have some numbers of orders of generalized coordinates and what is happening in this repository is that in the research generalized coordinates and order there's a model visualization folder. This is going to make visualizations of each of the model orders. So this is like a low order first order one and this is a sixth order and they're all put together in this collage that has all of the orders. Here it's 1 through eight but there's a app model block generator so that you can use any order you want. The outputs folder has one general case and then it separates into different orders. So we'll look at the uh third order generalized coordinates of motion model and there's a scenario generator that generates three different uh cases and each each situation for each order is going to get all these outputs. So first is the position uh position, velocity, acceleration, however many orders are being considered. What fraction of the time is the inference within the 95% confidence interval actually including the true value? Here's the dashboard. Here's the position, velocity, acceleration, and the free energy through time. the true and the inferred value here because it was essentially uh a smooth moving scenario. The third order approximation did really well. But if we look at that constant scenario with a first order approximation basically it just is a straight line. And if we look at the sign scenario here again the first order approximation all it can do is basically draw the best straight line through this whole data set. Whereas if we go to the sign with a third order approximation we're getting uh the ability to latch on to capture the position velocity acceleration. So coverage within the confidence interval dashboard, some information on the derivatives, the error profile, cumulative sum and free energy through time. Some of the artifacts at the very beginning and the very end are just relative to the the it being earlier late data points. uh mean squared error through time for each of those position, velocity, and acceleration. Uh histogram of the residuals, root mean squared for each of those states. Here, a scatter plot, the position, velocity, acceleration. So, it latched on pretty well. Again, if we look at the first order, true versus inferred here, all all it can do is just guess on this straight line uh a QQ plot. So, definitely leaves some room to be improved quantitatively, but just structurally was what I was seeking to get here. uh residuals through time, overall fit to the noisy data. Again, the goal was to go from noisy observations to the predictions of the underlying data, some metrics, uh written report and information on the scenario that is happening uh for each scenario that gets metaanalyzed across the scenarios. So here we have the mean correlation for these three scenarios the the constant the peace-wise mixed and the sign only. So let's just look at what the peacewise mixed looks like. This one has every 500 time steps it kind of switches into a different regime. So that really challenges the approximation and we can see that for example on the constant situation the all the orders do really well because it's pretty easy in the peace-wise and in the sign situation the low orders do bad and the higher orders do fine and there's diminishing returns once you get into these higher derivatives. Here's another way to look at this. This is the the 95% confidence interval showing how again in the two more complex situations the uh low order models are not able to work that well. But then increasingly high order models do better and better and similar patterns are observed for free energy values. Again, realistically at at super high orders, you're getting weird overfitting errors. And that is all happening from the uh the situation is is the car situation. It's just imagining that there's a car. It was based off of the hierarchical Gaussian filter. So here's the uh example that was uh based off of which was basically this hierarchical Gaussian filter situation. And in the generalized coordinates, it's configured through uh the ability to to uh run the suite across uh different orders. in SRC is where the model is. So here's the basis of the RX infer model and here's the constraint block that is used to generate all those combinations. So again, generalized coordinates. This uh folder, this one is just three orders. This one's just uh uh three different uh orders of of motion. And then I generalized that into the n orders of generalized coordinates in that folder. hgf is if you want to look more at how the hierarchical gaussian filter works and all that is based upon these rx infer examples. And also while you're here check out the visualization methods folder also in research. This is a generalized way to look at all the different RX infer models. So whether it's the the coin toss or it's the drone model or this generalized coordinates model the the base form or it's a hidden marov model.

These are some visualization methods. So, hope that is useful and interesting for those who like open-source statistical methods, RX and for learning about generalized coordinates, free energy principle, pathbased formulations, rough paths, all the good stuff. So, thanks for watching. Bye.